Module II

Blockchain - Definition, architecture, elements of blockchain, benefits and limitations, types of blockchain,. Consensus - definition, types, consensus in blockchain.

Decentralization - Decentralization using blockchain, Methods of decentralization, Routes to decentralization, Blockchain and full ecosystem decentralization.

Blockchain

Base - A paper entitled “Bitcoin: A Peer-to-Peer Electronic Cash System” written on the topic of peer-to-peer electronic cash under the pseudonym Satoshi Nakamoto.

It introduced the term chain of blocks.

The term chain of blocks evolved over the years into the word blockchain.

Definition: Blockchain is a peer-to-peer, distributed ledger that is cryptographically-secure, append-only, immutable (extremely hard to change), and updateable only via consensus or agreement among peers.

1. Peer-to-peer

There is no central controller in the network, and all participants talk to each other directly.

This property allows for cash transactions to be exchanged directly among the peers without a third-party involvement, such as by a bank.

2. Distributed ledger

A ledger is spread across the network among all peers in the network, and each peer holds a copy of the complete ledger.

3. Cryptographically-secure

Cryptography has been used to provide security services which make this ledger secure against tampering and misuse.

These services include non-repudiation, data integrity, and data origin authentication.

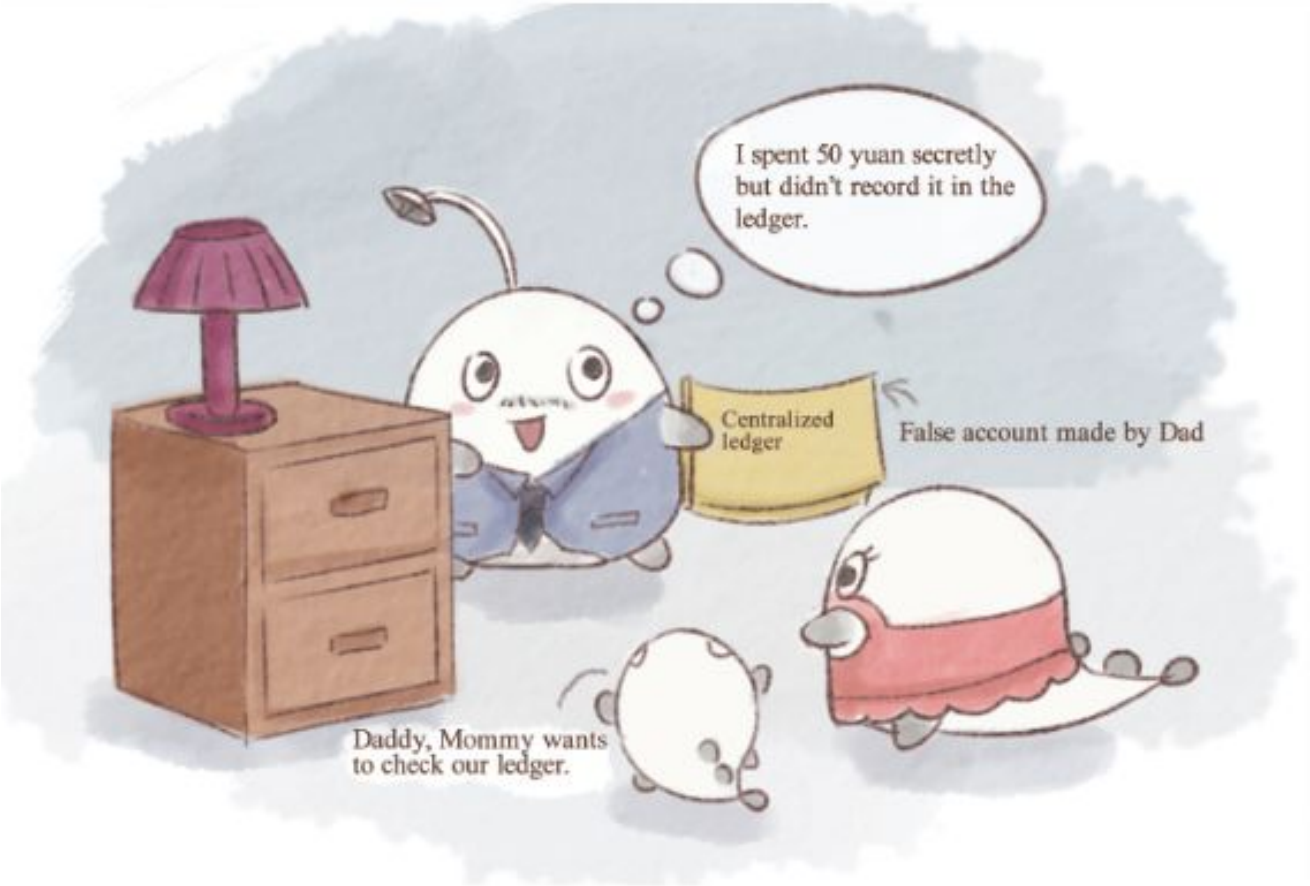


Fig. 2-3: Centralized family ledger.

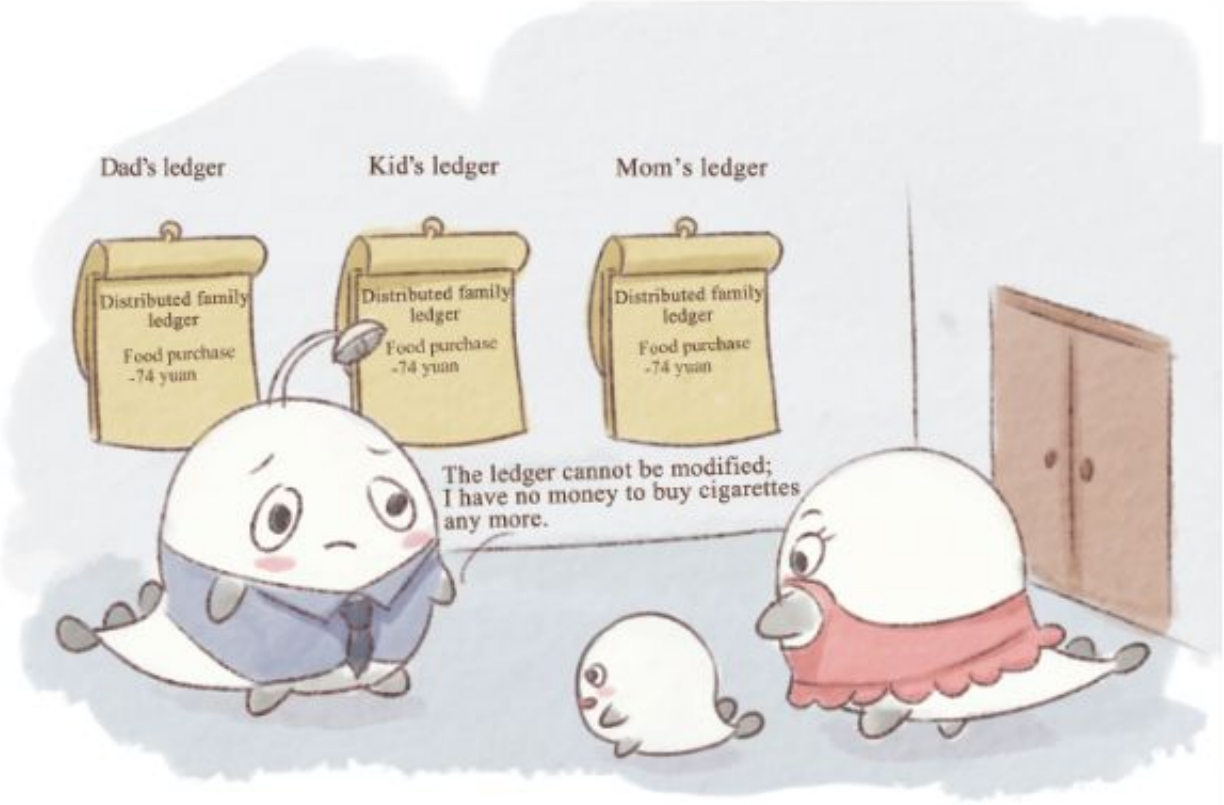
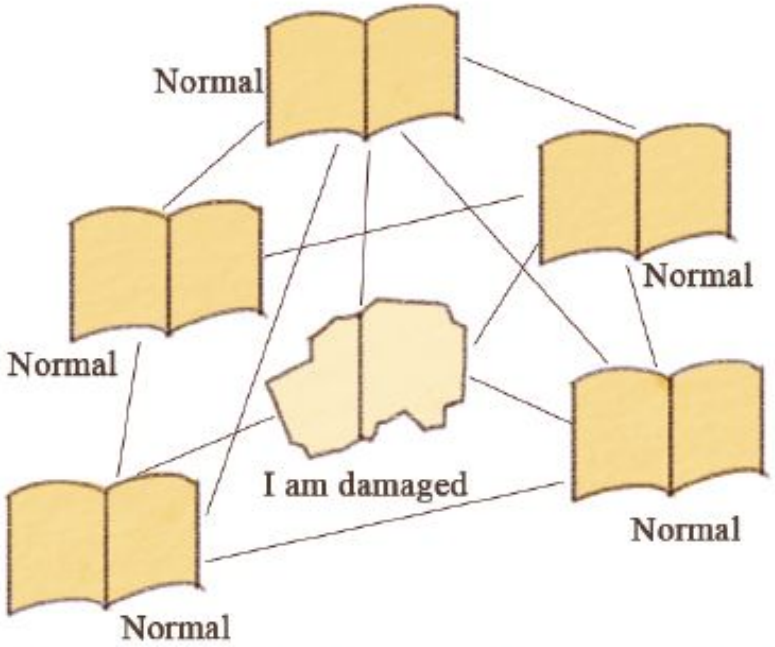
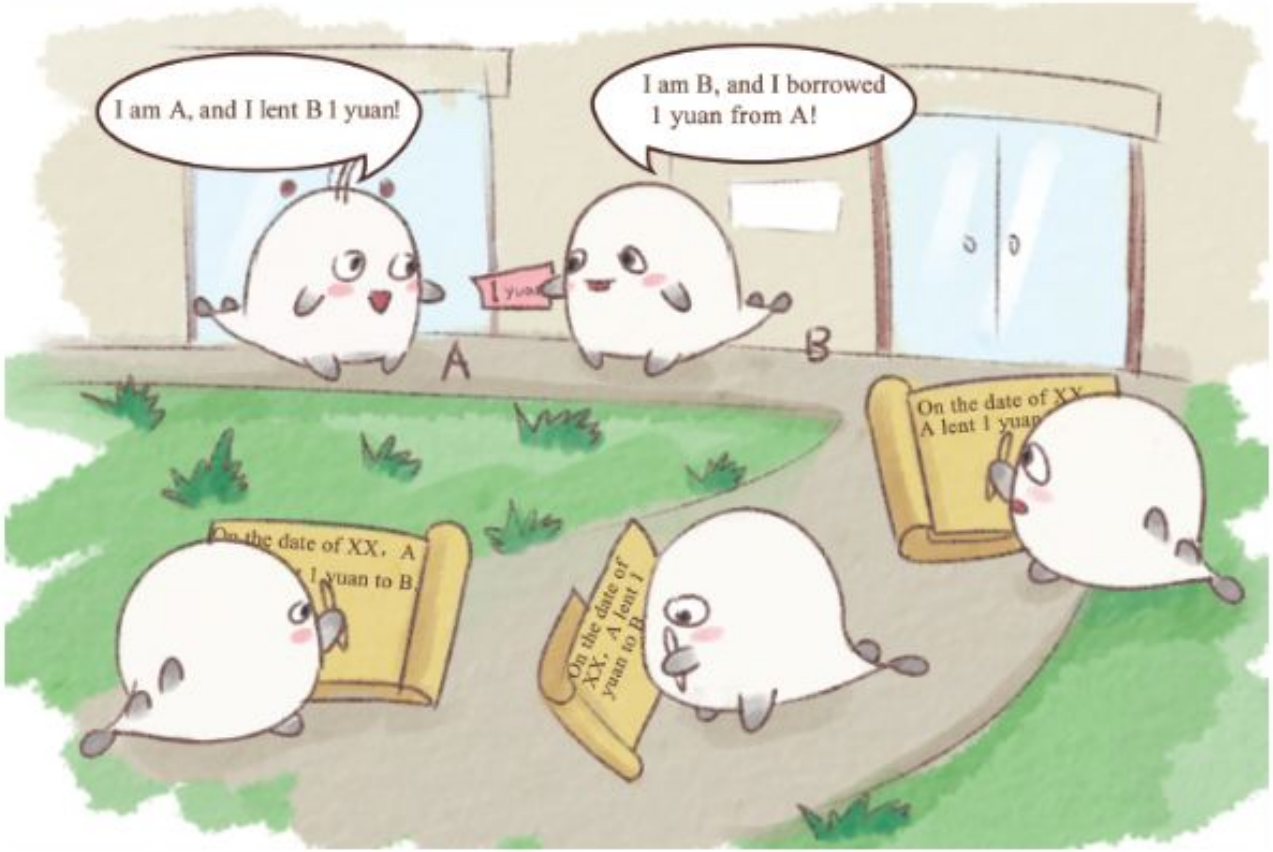


Fig. 2-4: Distributed family ledger.

Decentralization



If a single ledger is damaged, the system can still proceed normally.



Bookkeeping in Decentralized way

4. Append-only

Data can only be added to the blockchain in time-ordered sequential order.

Once data is added to the blockchain, it is almost impossible to change that data and can be considered practically immutable.

5. Updateable via consensus

There is no central authority in control of updating the ledger.

Instead, any update made to the blockchain is validated against strict criteria defined by the blockchain protocol and added to the blockchain only after a consensus has been reached among all participating peers/nodes on the network.

To achieve consensus, there are various consensus facilitation algorithms which ensure that all parties are in agreement.

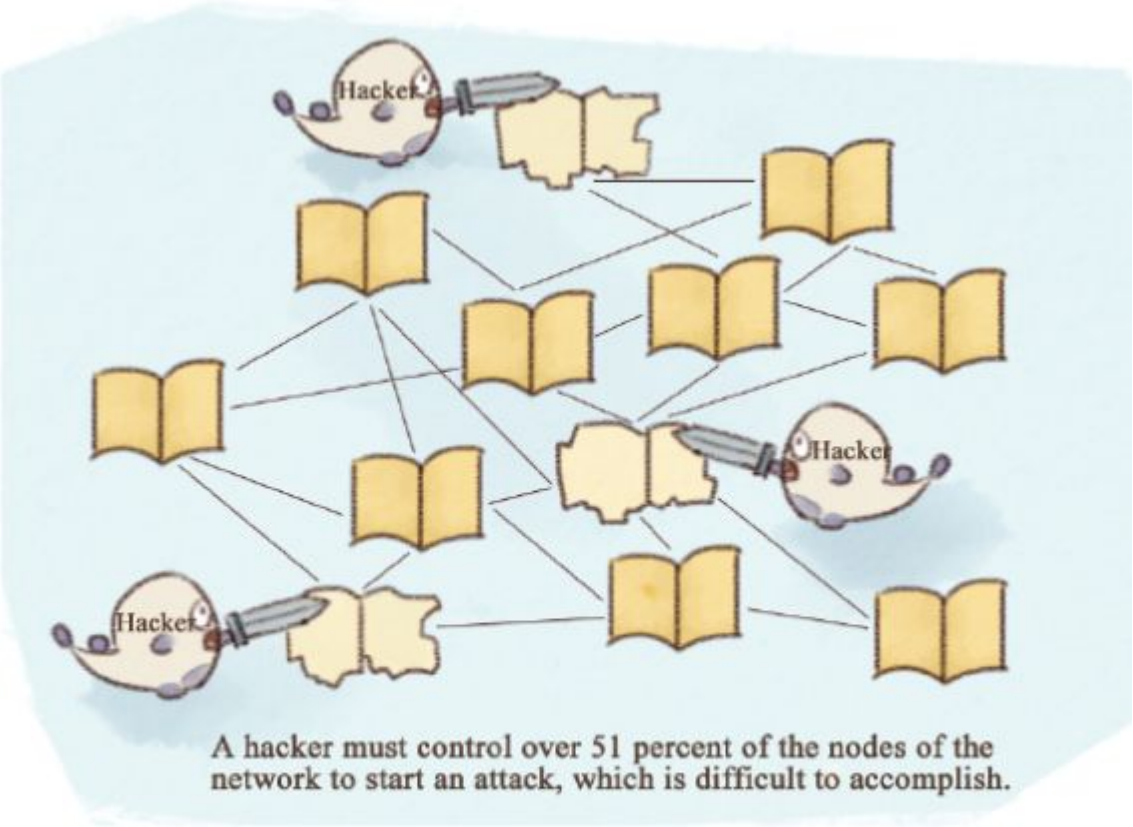
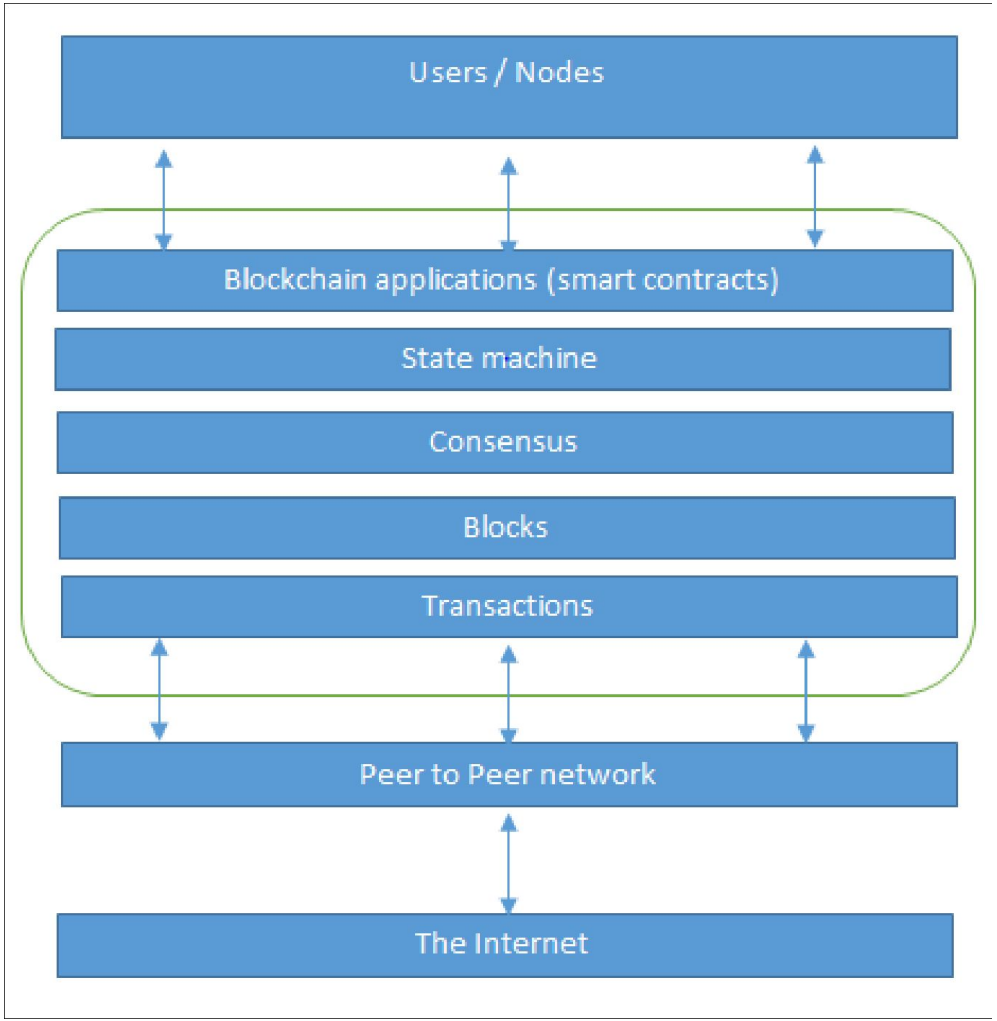


Fig. 2-66: The ledger cannot be compromised.

Network view of a blockchain

Blockchain can be thought of as a layer of a distributed peer-to-peer network running on top of the internet, analogous to SMTP, HTTP, or FTP running on top of TCP/IP.


Users / Nodes
Blockchain applications (smart contracts)
State machine
Consensus
Blocks
Transactions
Peer to Peer network
The Internet

Generic Structure of a block

A block is merely a selection of transactions bundled together and organized logically.

A transaction is a record of an event, for example, the event of transferring cash from a sender's account to a beneficiary's account.

A block is made up of transactions, and its size varies depending on the type and design of the blockchain in use.

A reference to a previous block is also included in the block unless it is a genesis block.

A genesis block is the first block in the blockchain that is hardcoded at the time the blockchain was first started.

A nonce (number used only once) is a number that is generated and used only once.

A nonce is used extensively in many cryptographic operations to provide replay protection, authentication, and encryption.

In blockchain, it's used in PoW consensus algorithms and for transaction replay protection.

Merkle root is a hash of all of the nodes of a Merkle tree.

In the blockchain world, Merkle trees are commonly used to allow efficient verification of transactions.

Merkle root in a blockchain is present in the block header section of a block, which is the hash of all transactions in a block.

This means that verifying only the Merkle root is required to verify all transactions present in the Merkle tree instead of verifying all transactions one by one.

Block Structure in short

The block is a record that contains the transaction data details. It comprises of the following details:

1) Hash of the block – Alphanumeric number to identify the block

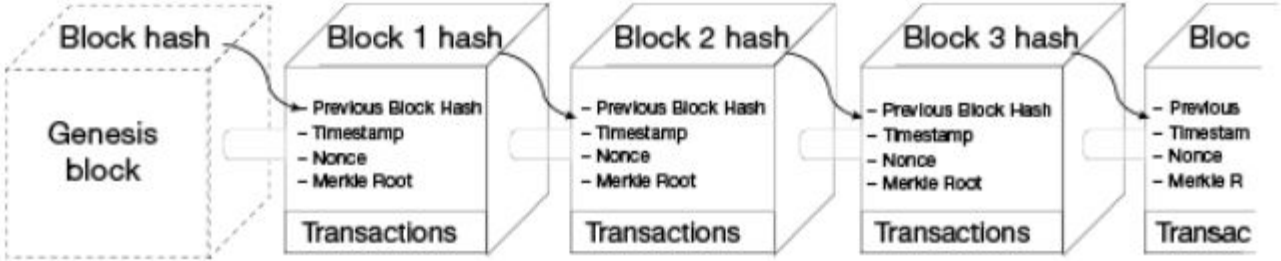
2) Hash of the previous block

3) Timestamp

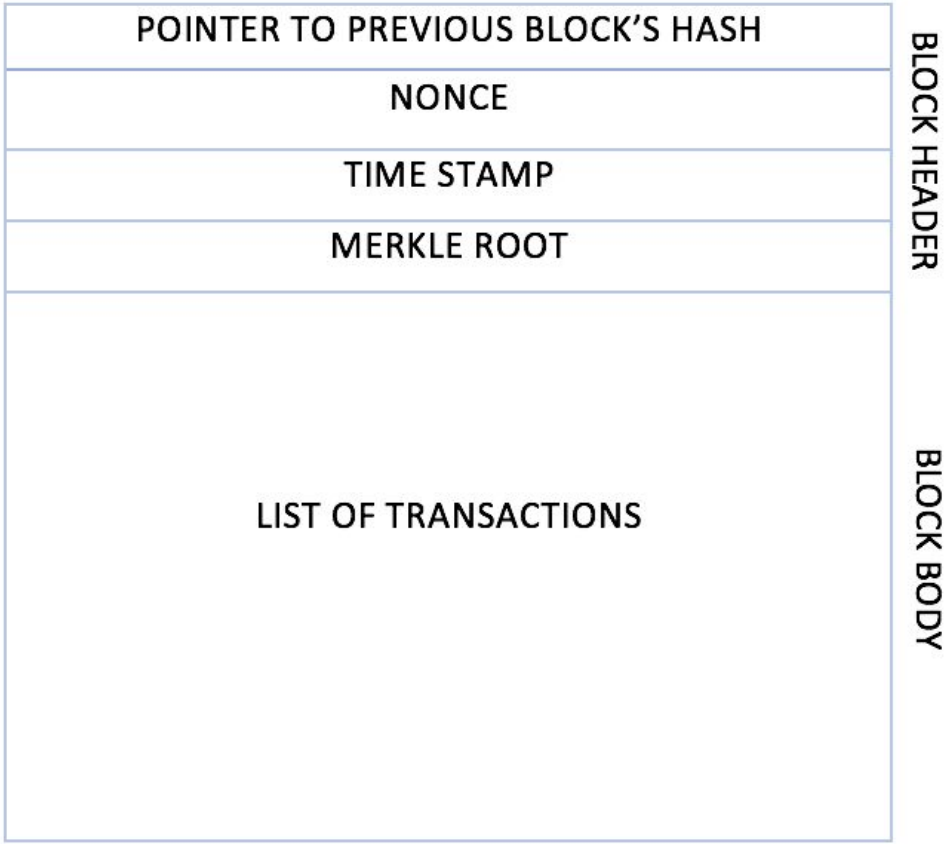
4) Nonce – the random number used to vary the value of the hash

5) Merkle root – hash of all the hashes of all the transactions in the block

6) Transaction data. This contains details of several transactions



Generic Structure of a Block



In Bitcoin: A block contains more than 500 transactions on average.

Average block size: 1 MB

Larger blocks can process large no. of transactions but longer time for verification and propagation

Generic elements of a blockchain

1. Address: Addresses are unique identifiers used in a blockchain transaction to denote senders and recipients.

An address is usually a public key or derived from a public key.

Addresses can be reused by the same user.

In practice, however, a single user may not use the same address again and generate a new one for each transaction.

This newly-created address will be unique.

A good practice is for users to generate a new address for each transaction in order to avoid linking transactions to the common owner, thus preventing identification.

2. Transaction: A transaction is the fundamental unit of a blockchain.

A transaction represents a transfer of value from one address to another.

3. Block: A block is composed of multiple transactions and other elements, such as the previous block hash (hash pointer), timestamp, and nonce.

4. Peer-to-peer network: As the name implies, a peer-to-peer network is a network topology wherein all peers can communicate with each other and send and receive messages.

5. Scripting or programming language: Scripts or programs perform various operations on a transaction in order to facilitate various functions.

For example, in Bitcoin, transaction scripts are predefined in a language called Script, which consist of sets of commands that allow nodes to transfer tokens from one address to another.

Script is a limited language, it only allows essential operations that are necessary for executing transactions, but it does not allow for arbitrary program development.

As such, Bitcoin script language cannot be called Turing complete.

In simple words, Turing complete language means that it can perform any computation.

Turing complete languages need loops and branching capability to perform complex computations.

Therefore, Bitcoin's scripting language is not Turing complete, whereas Ethereum's Solidity language is.

To facilitate arbitrary program development on a blockchain, Turing complete programming language is needed, and it is now a very desirable feature of blockchains.

6. Virtual machine: A virtual machine allows Turing complete code to be run on a blockchain (as smart contracts);

However, virtual machines are not available on all blockchains.

Various blockchains use virtual machines to run programs such as Ethereum Virtual Machine (EVM) and Chain Virtual Machine (CVM).

EVM is used in Ethereum blockchain.

CVM is a virtual machine developed for and used in an enterprise-grade blockchain called Chain Core.

7. State machine: A blockchain can be viewed as a state transition mechanism whereby a state is modified from its initial form to the next one and eventually to a final form by nodes on the blockchain network as a result of a transaction execution, validation, and finalization process.

8. Node: A node in a blockchain network performs various functions depending on its role.

A node can propose and validate transactions and perform mining to facilitate consensus and secure the blockchain.

This goal is achieved by following a consensus protocol (most commonly PoW).

Nodes can also perform other functions such as simple payment verification (lightweight nodes), validation, and many other functions depending on the type of the blockchain used and the role assigned to the node.

Nodes also perform a transaction signing function.

Transactions are first created by nodes and then also digitally signed by nodes using private keys as proof that they are the legitimate owner of the asset that they wish to transfer to someone else on the blockchain network.

This asset is usually a token or virtual currency, such as Bitcoin, but it can also be any real-world asset represented on the blockchain by using tokens.

9. Smart contract: These programs run on top of the blockchain and encapsulate the business logic to be executed when certain conditions are met.

These programs are enforceable and automatically executable.

The smart contract feature is not available on all blockchain platforms.

Smart contracts have many use cases, including but not limited to identity management, capital markets, trade finance, record management, insurance, and e-governance

Benefits and limitations of blockchain

Benefits

i. Decentralization:

- Core concept and benefit of the blockchain.
- There is no need for a trusted third party or intermediary to validate transactions; instead, a consensus mechanism is used to agree on the validity of transactions.

ii. Transparency and trust:

- Blockchains are shared and everyone can see what is on the blockchain, this allows the system to be transparent.

iii. Immutability:

Once the data has been written to the blockchain, it is extremely difficult to change it back.

iv. High availability:

- As the system is based on thousands of nodes in a peer-to-peer network, and the data is replicated and updated on every node, the system becomes highly available.
- Even if some nodes leave the network or become inaccessible, the network as a whole continues to work, thus making it highly available.
- This redundancy results in high availability.

v. Highly secure:

- All transactions on a blockchain are cryptographically secured and thus provide network integrity.

vi. Simplification of current paradigms:

A blockchain can serve as a single shared ledger among many interested parties, this can result in simplifying the model by reducing the complexity of managing the separate systems maintained by each entity.

vii. Faster dealings:

In the financial industry, especially in post-trade settlement functions, blockchain can play a vital role by enabling the quick settlement of trades.

Blockchain does not require a lengthy process of verification, reconciliation, and clearance because a single version of agreed-upon data is already available on a shared ledger between financial organizations.

viii. Cost saving:

As no trusted third party or clearing house is required in the blockchain model, this can massively eliminate overhead costs in the form of the fees which are paid to such parties.

Challenges/Limitations of Blockchain

The most sensitive blockchain problems are:

- Scalability
- Adaptability
- Regulation
- Relatively immature technology
- Privacy

Tiers of blockchain technology

Blockchain 1.0:

This tier was introduced with the invention of Bitcoin.

Primarily used for cryptocurrencies.

Also, as Bitcoin was the first implementation of cryptocurrencies.

This first generation of blockchain technology includes only cryptographic currencies.

All alternative cryptocurrencies as well as Bitcoin fall into this category.

This generation started in 2009.

Blockchain 2.0:

This second blockchain generation is used by financial services and smart contracts.

This tier includes various financial assets, such as derivatives, options, swaps, and bonds. Applications that go beyond currency, finance, and markets are incorporated at this tier.

Ethereum, Hyperledger, and other newer blockchain platforms are considered part of Blockchain 2.0.

This generation started when ideas related to using blockchain for other purposes started to emerge in 2010.

Blockchain 3.0:

This third blockchain generation is used to implement applications beyond the financial services industry and is used in government, health, media, the arts, and justice.

As in Blockchain 2.0, Ethereum, Hyperledger, and newer blockchains with the ability to code smart contracts are considered part of this blockchain technology tier.

This generation of blockchain emerged around 2012 when multiple applications of blockchain technology in different industries were researched.

Blockchain X.0:

This generation represents a vision of blockchain singularity where one day there will be a public blockchain service available that anyone can use just like the Google search engine.

It will provide services for all realms of society.

It will be a public and open distributed ledger with general-purpose rational agents running on a blockchain, making decisions, and interacting with other intelligent autonomous agents on behalf of people, and regulated by code instead of law or paper contracts.

This does not mean that law and contracts will disappear, instead law and contracts will be implementable in code.

Features of a blockchain

Distributed consensus:

This mechanism allows a blockchain to present a single version of the truth, which is agreed upon by all parties without the requirement of a central authority.

Transaction verification:

Any transactions posted from the nodes on the blockchain are verified based on a predetermined set of rules.

Only valid transactions are selected for inclusion in a block.

Platform for smart contracts:

A blockchain is a platform on which programs can run to execute business logic on behalf of the users.

Not all blockchains have a mechanism to execute smart contracts; however, this is a very desirable feature.

Transferring value between peers:

Blockchain enables the transfer of value between its users via tokens.

Tokens can be thought of as a carrier of value.

Generation of cryptocurrency:

This feature is optional depending on the type of blockchain in use.

A blockchain can create cryptocurrency as an incentive to its miners who validate the transactions and spend resources to secure the blockchain.

Smart property:

It is now possible to link a digital or physical asset to the blockchain in such a secure and precise manner that it cannot be claimed by anyone else.

You are in full control of your asset, and it cannot be double-spent or double-owned.

Compare this with a digital music file, for example, which can be copied many times without any controls.

blockchain based Digital Rights Management (DRM) schemes can enforce the Digital Rights Management fully.

Provider of security:

The blockchain is based on proven cryptographic technology that ensures the integrity and availability of data.

Generally, confidentiality is not provided due to the requirements of transparency.

Other security services, such as non-repudiation and authentication, are also provided by blockchain, as all actions are secured using private keys and digital signatures.

Immutability:

Once records are added to the blockchain, they are immutable.

There is the remote possibility of rolling back changes, but this is to be avoided at all costs as doing so would consume an exorbitant amount of computing resources.

Uniqueness:

This blockchain feature ensures that every transaction is unique and has not already been spent (double-spend problem).

This feature is especially relevant with cryptocurrencies, where detection and avoidance of double spending are a vital requirement.

Types of blockchain

Based on technical and business usage perspective.

1. Distributed ledgers

All blockchains technically fall under the umbrella of distributed ledgers.

Critical difference between a distributed ledger and blockchain is that a distributed ledger does not consist of blocks of transactions.

The records(transactions) are stored contiguously instead of being sorted into blocks.

Blockchain is a special type of distributed ledger that is comprised of blocks of transactions.

eg: distributed ledger that does not use blocks of transactions is R3's Corda.

Bitcoin and Ethereum make use of blocks to update the shared database.

2. Distributed Ledger Technology

Distributed ledger is distributed among its participants and spread across multiple sites or organizations.

It can be either private or public

The records are stored contiguously instead of being sorted into blocks.

3. Public blockchains

Public blockchains are not owned by anyone.

They are open to the public, and anyone can participate as a node in the decision-making process.

Users may or may not be rewarded for their participation.

All users of these prmissionless or unpermissioned ledgers maintain a copy of the ledger on their local nodes and use a distributed consensus mechanism to decide the eventual state of the ledger.

Bitcoin and Ethereum are both considered public blockchains.

4. Private blockchains

Open only to a consortium or group of individuals or organizations who have decided to share the ledger among themselves.

examples: HydraChain and Quorum.

Optionally, both of these blockchains can also run in public mode if required.

5. Semiprivate blockchains

Part of the blockchain is private and part of it is public.

Still just a concept today.

The private part is controlled by a group of individuals, while the public part is open for participation by anyone.

Can also be called a semi-decentralized model.

6. Sidechains

More precisely known as pegged sidechains.

A concept whereby coins can be moved from one blockchain to another and moved back again.

Typical uses include the creation of new altcoins (alternative cryptocurrencies) whereby coins are burnt as a proof of an adequate stake.

Burnt or burning the coins in this context means that the coins are sent to an address which is unspendable and this process makes the burnt coins irrecoverable.

This mechanism is used to bootstrap a new currency or introduce scarcity which results in increased value of the coin.

Rootstock is an example of a sidechain

7. Permissioned ledger

Is a blockchain where participants of the network are already known and trusted.

Permissioned ledgers do not need to use a distributed consensus mechanism;

instead, an agreement protocol is used to maintain a shared version of the truth about the state of the records on the blockchain

In this case, for verification of transactions on the chain, all verifiers are already preselected by a central authority and typically there is no need for a mining mechanism.

There is also no requirement for a permissioned blockchain to be private.

It can be a public blockchain but with regulated access control.

For example, Bitcoin can become a permissioned ledger if an access control layer is introduced on top of it that verifies the identity of a user and then allows access to the blockchain.

8. Shared ledger

This is a generic term used to describe any application or database that is shared by the public or a consortium.

Generally, all blockchains, fall into the category of a shared ledger.

9. Fully private and proprietary blockchains

There is no mainstream application of these types of blockchains, as they deviate from the core concept of decentralization in blockchain technology.

Nonetheless, in specific private settings within an organization, there could be a need to share data and provide some level of guarantee of the authenticity of the data.

An example of this type of blockchain might be to allow for collaboration and the sharing data between various government departments.

In that case, no complex consensus mechanism is required, apart from simple state machine replication and an agreement protocol with known central validators.

10. Tokenized blockchains

These blockchains are standard blockchains that generate cryptocurrency as a result of a consensus process via mining or initial distribution.

Bitcoin and Ethereum are prime examples of this type of blockchain.

Tokenless blockchains

They do not have the basic unit for the transfer of value.

However, they are still valuable in situations where there is no need to transfer value between nodes and only the sharing of data among various trusted parties is required.

This is similar to full private blockchains, the only difference being that use of tokens is not required.

This can also be thought of as a shared distributed ledger used for storing data.

It has benefits in terms of immutability, security, and consensus driven updates.

Not used for common blockchain application of value transfer or cryptocurrency.

How blockchain works

Nodes are either miners or block signers

- Miners: create new blocks and mint cryptocurrency (coins)
- Block signers: validates and digitally sign the transactions.

A critical decision that every blockchain network has to make is to figure out that which node will append the next block to the blockchain.

This decision is made using a consensus mechanism.

How blockchain accumulates blocks

Step 1: A node starts a transaction by first creating and then digitally signing it with its private key.

A transaction can represent various actions in a blockchain.

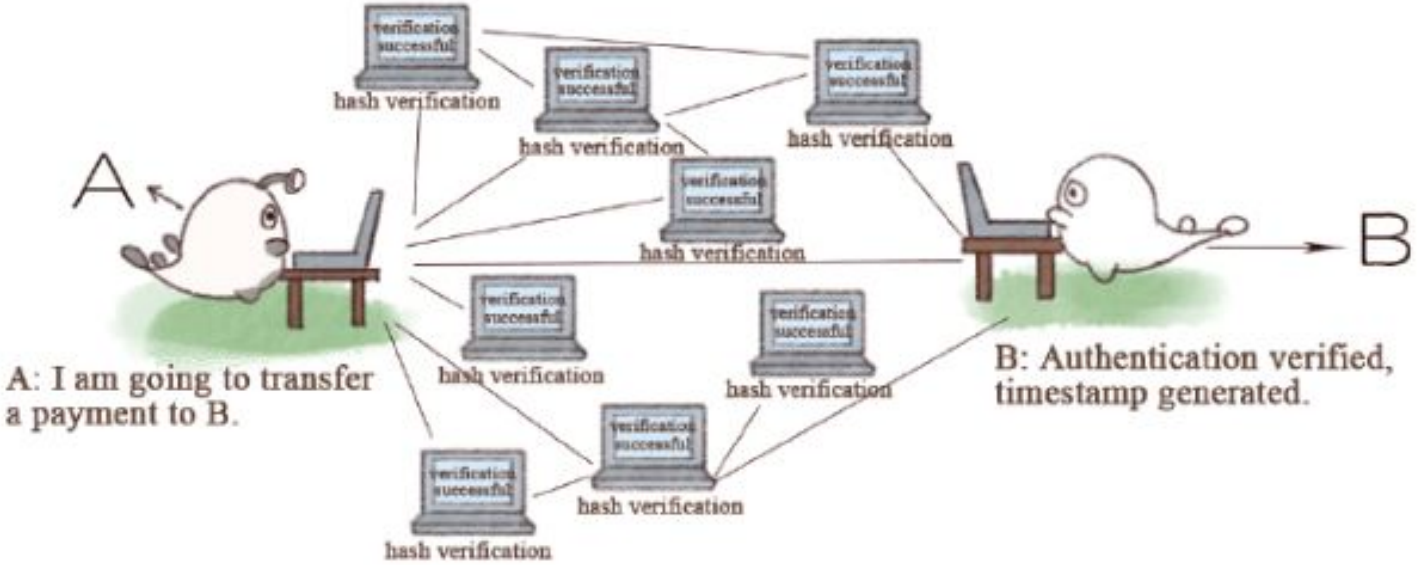
Most commonly this is a data structure that represents transfer of value between users on the blockchain network.

Transaction data structure usually consists of some logic of transfer of value, relevant rules, source and destination addresses, and other validation information.

Step 2: A transaction is propagated (flooded) by using a flooding protocol, called Gossip protocol, to peers that validate the transaction based on preset criteria.

Usually, more than one node are required to verify the transaction.

Broadcasting to network



Step 3: Once the transaction is validated, it is included in a block, which is then propagated onto the network.

At this point, the transaction is considered confirmed.

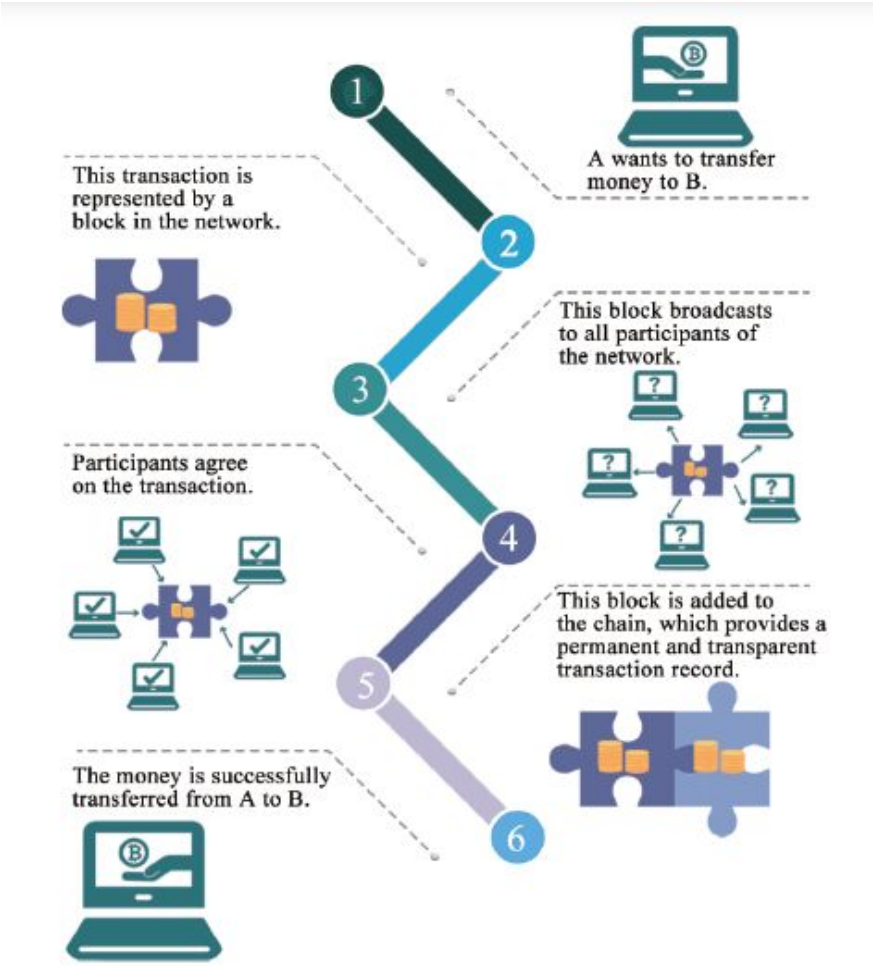
Step 4: The newly-created block now becomes part of the ledger, and the next block links itself cryptographically back to this block.

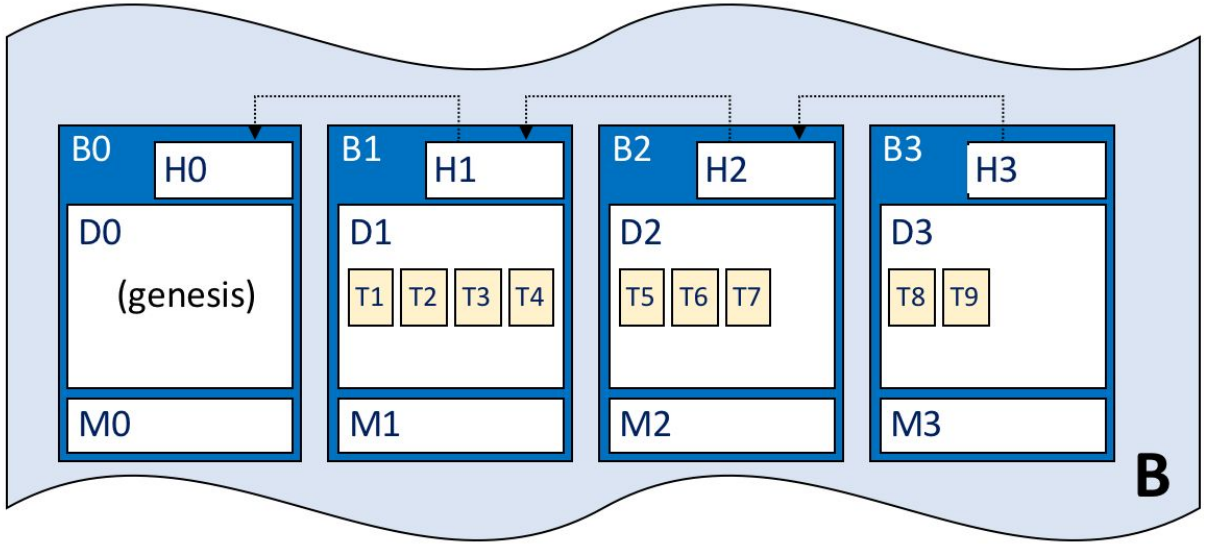
This link is a hash pointer.

At this stage, the transaction gets its second confirmation and the block gets its first confirmation.

Step 5: Transactions are then reconfirmed every time a new block is created.

Steps 4 and 5 are considered non-compulsory, as the transaction itself is finalized in step 3; however, block confirmation and further transaction re-confirmations, if required, are then carried out in step 4 and step 5


1
A wants to transfer money to B.
2
This transaction is represented by a block in the network.
3
This block broadcasts to all participants of the network.
4
Participants agree on the transaction.
5
This block is added to the chain, which provides a permanent and transparent transaction record.
6
The money is successfully transferred from A to B.



B	Blockchain
B1	Block
H3	Block header
D1	Block data
T5	Transaction
M3	Block metadata
H1 H2	H2 is chained to H1

Consensus

Consensus is a process of agreement between distrusting nodes on the final state of data.

To achieve consensus, different algorithms are used.

The choice of the consensus algorithm is governed by the type of blockchain in use.

The process of attaining agreement on common state or value among multiple nodes despite the failure of some nodes is known as distributed consensus.

Consensus mechanism

A consensus mechanism is a set of steps that are taken by most or all nodes in a blockchain to agree on a proposed state or value.

The Consensus Problem

Which one would be the next block?

Unconfirmed TX

TX16
TX17
TX87
TX49
TX37

Miner 1

Unconfirmed TX

TX17
TX22
TX87
TX37
TX88

Miner 2

Unconfirmed TX

TX16
TX17
TX22
TX31

Miner 3

NPTEL

Mining

Mining is a process by which new blocks are added to the blockchain.

Blocks contain transactions that are validated via the mining process by mining nodes on the Bitcoin network.

Blocks, once mined and verified are added to the blockchain which keeps the blockchain growing.

This process is resource-intensive

Roughly one new block is created (mined) every 10 minutes to control the frequency of generation of bitcoins.

Approximately 144 blocks, that is, 1,728 bitcoins are generated per day.

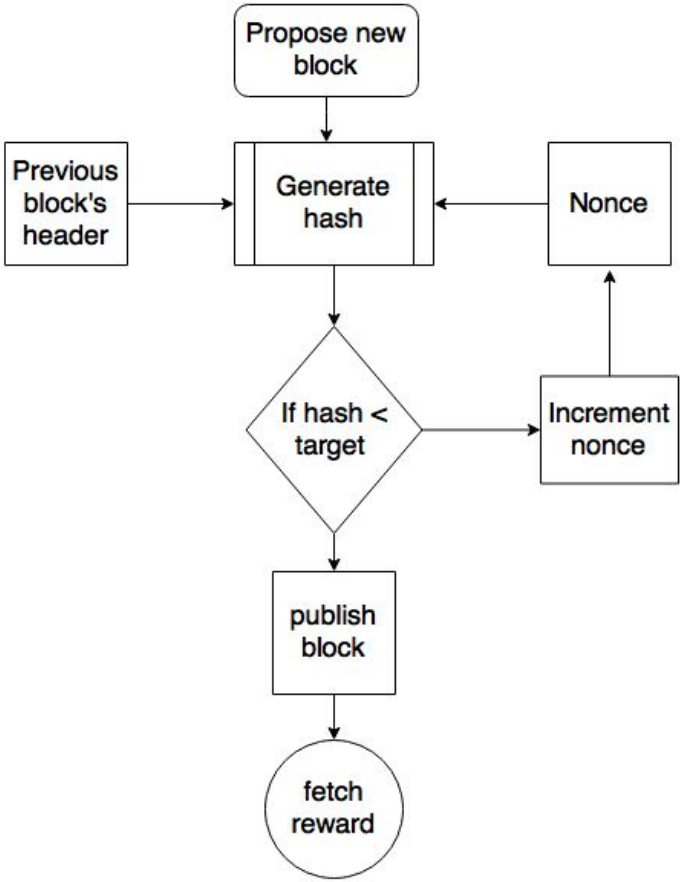
The rate of creation of new bitcoins decreases by 50%, every 210,000 blocks, roughly every 4 years.

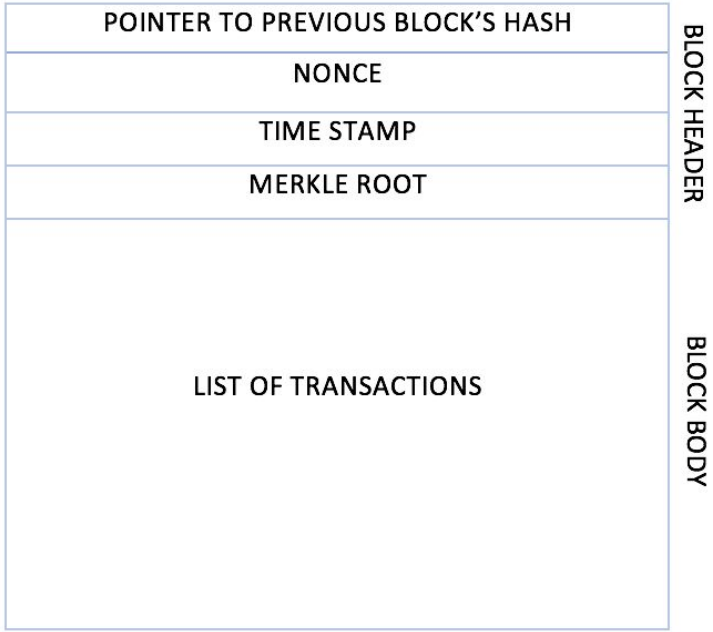
When bitcoin was initially introduced, the block reward was 50 bitcoins;

then in 2012, this was reduced to 25 bitcoins.

In July 2016, this was further reduced to 12.5 coins (12 coins)

on July 4, 2020, this was reduced to down to approximately six coins.





Mining Process in PoW

Requirements for a consensus mechanism

Agreement: All honest nodes decide on the same value.

Termination: All honest nodes terminate execution of the consensus process and eventually reach a decision.

Validity: The value agreed upon by all honest nodes must be the same as the initial value proposed by at least one honest node

Fault tolerant: The consensus algorithm should be able to run in the presence of faulty or malicious nodes (Byzantine nodes).

Integrity: This is a requirement that no node can make the decision more than once in a single consensus cycle.

Types of consensus mechanisms

There are two general categories of consensus mechanisms.

Traditional Byzantine Fault Tolerance (BFT)-based: With no compute-intensive operations, this method relies on a simple scheme of nodes that are publisher-signed messages.

Eventually,when a certain number of messages are received, then an agreement is reached.

Leader election-based consensus mechanisms: This arrangement requires nodes to compete in a leader-election lottery, and the node that wins proposes a final value.

For example, the PoW used in Bitcoin falls into this category.

Consensus in blockchain

Two categories exist,

Proof-based, leader-election lottery based, or the Nakamoto consensus whereby a leader is elected at random (using an algorithm) and proposes a final value.

This category is also referred to as the fully decentralized or permissionless type of consensus mechanism.

This type is well used in the Bitcoin and Ethereum blockchain in the form of a PoW mechanism.

BFT-based is a more traditional approach based on rounds of votes.

This class of consensus is also known as the consortium or permissioned type of consensus mechanism.

BFT-based consensus mechanisms perform well when there are a limited number of nodes, but they do not scale well.

On the other hand, leader-election lottery based (PoW) type consensus mechanisms scale very well but perform very slowly.

As there is significant research being conducted in this area, new types of consensus mechanism are also emerging, such as the semi-decentralized type, which is used in the Ripple network.

Some of the major consensus algorithms available today are discussed below.

Proof of Work (PoW): This type of consensus mechanism relies on proof that adequate computational resources have been spent before proposing a value for acceptance by the network.

This scheme is used in Bitcoin, Litecoin, and other cryptocurrency blockchains.

Proof of Stake (PoS): This algorithm works on the idea that a node or user has an adequate stake in the system;

that is, the user has invested enough in the system so that any malicious attempt by that user would outweigh the benefits of performing such an attack on the network.

This idea was first introduced by Peercoin, and it is going to be used in the Ethereum blockchain version called Serenity.

Another important concept in PoS is coin age, which is a criterion derived from the amount of time and number of coins that have not been spent.

In this model, the chances of proposing and signing the next block increase with the coin age.

Delegated Proof of Stake (DPoS): This is an innovation over standard PoS, whereby each node that has a stake in the system can delegate the validation of a transaction to other nodes by voting.

It is used in the BitShares blockchain.

Proof of Elapsed Time (PoET): Introduced by Intel in 2016, PoET uses a Trusted Execution Environment (TEE) to provide randomness and safety in the leader election process via a guaranteed wait time.

It requires the Intel Software Guard Extensions (SGX) processor to provide the security guarantee for it to be secure.

Proof of Deposit (PoD): In this case, nodes that wish to participate in the network have to make a security deposit before they can mine and propose blocks.

This mechanism is used in the Tendermint blockchain.

Proof of Importance (PoI): Derived from PoS.

PoI not only relies on how large a stake a user has in the system, but it also monitors the usage and movement of tokens by the user in order to establish a level of trust and importance.

It is used in the NEM coin blockchain. (https://nem.io/)

Federated consensus or federated Byzantine consensus: This mechanism is used

in the stellar consensus protocol.

Nodes in this protocol retain a group of publicly-trusted peers and propagate only those transactions that have been validated by the majority of trusted nodes.

Reputation-based mechanisms: As the name suggests, a leader is elected by the reputation it has built over time on the network.

It is based on the votes of other members.

Proof of Activity (PoA): This scheme is a combination of PoS and PoW, which ensures that a stakeholder is selected in a pseudorandom but uniform fashion.

This is a comparatively more energy-efficient mechanism as compared to PoW.

It utilizes a new concept called Follow the Satoshi.

In this scheme, PoW and PoS are combined together to achieve consensus and good level of security.

This scheme is more energy efficient as PoW is used only in the first stage of the mechanism, after the first stage it switches to PoS which consumes negligible energy.

Proof of Capacity (PoC): This scheme uses hard disk space as a resource to mine the blocks.

This is different from PoW, where CPU resources are used.

In PoC, hard disk space is utilized for mining and as such is also known as hard drive mining.

This concept was first introduced in the Burstcoin cryptocurrency.

Proof of Storage (PoS): This scheme allows for the outsourcing of storage capacity.

This scheme is based on the concept that a particular piece of data is probably stored by a node which serves as a means to participate in the consensus mechanism.

Several variations of this scheme have been proposed, such as Proof of Replication, Proof of Data Possession, Proof of Space, and Proof of Space-Time.

Decentralization using blockchain

Decentralization is the core benefit and service provided by blockchain technology.

By design, blockchain model uses decentralization.

Allows anyone to compete to become the decision-making authority.

This competition is governed by a consensus mechanism, and the most commonly used method is known as Proof of Work (PoW).

Centralized systems

Conventional (client-server) IT systems in which there is a single authority that controls the system, and who is solely in charge of all operations on the system.

All users of a centralized system are dependent on a single source of service.

The majority of online service providers including Google, Amazon, eBay, Apple's App Store, and others use this conventional model for delivering services.

Distributed system

Data and computation are spread across multiple nodes in the network.

The critical difference between a decentralized system and distributed system is that in a distributed system, there still exists a central authority that governs the entire system; whereas, in a decentralized system, no such authority exists.

Decentralized system

A decentralized system is a type of network where nodes are not dependent on a single master node; instead, control is distributed among many nodes.

A significant innovation in the decentralized paradigm that has given rise to this new era of decentralization of applications is decentralized consensus.

This mechanism came into play with Bitcoin, and it enables a user to agree on something via a consensus algorithm without the need for a central, trusted third party, intermediary, or service provider.

Methods of decentralization

Two methods:

disintermediation and competition (Contest-driven decentralization).

Disintermediation

- Send money(or anything) directly to the recipient.
- No bank/ intermediary is involved.
- Address of the recipient in blockchain is to be known.
- This model can be used not only in finance but in many different industries as well

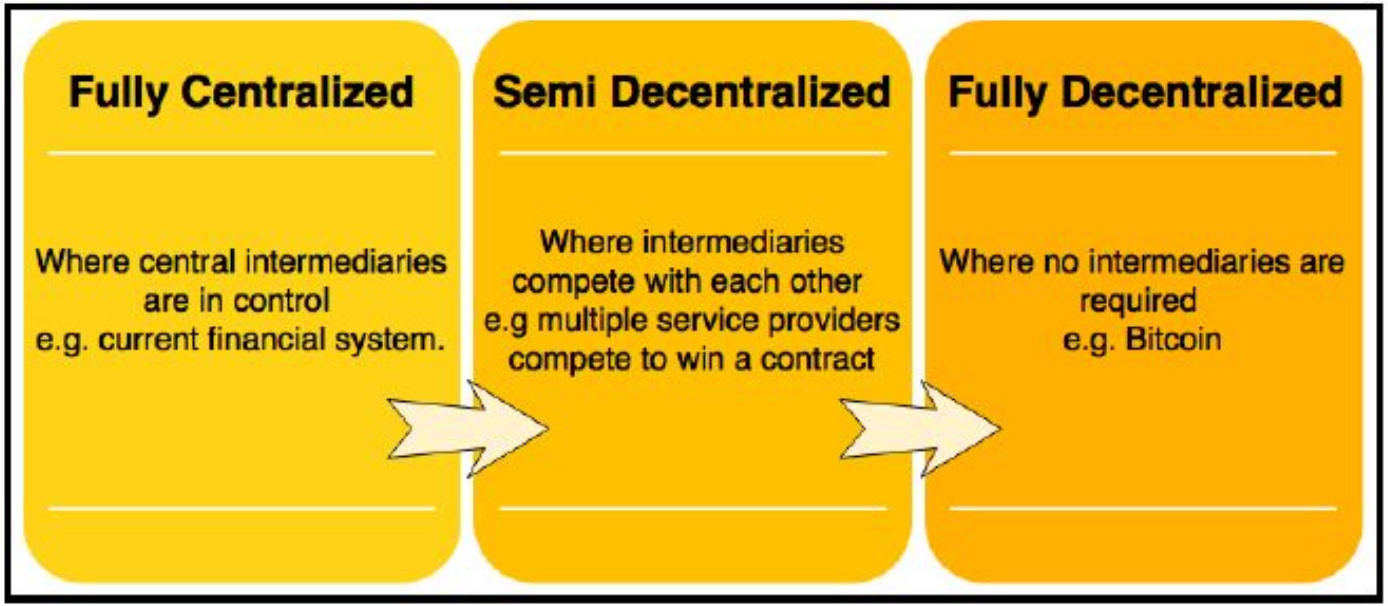
Contest-driven decentralization

In the method involving competition, different service providers compete with each other in order to be selected for the provision of services by the system.

This paradigm does not achieve complete decentralization.

However, to a certain degree, it ensures that an intermediary or service provider is not monopolizing the service.

In the context of blockchain technology, a system can choose an external data provider from a large number of providers based on their reputation, previous score, reviews, and quality of service.



Scale of Decentralization

Routes to decentralization

Many initiatives are now being used for achieving decentralization.

The Bitcoin blockchain is typically the first choice for many.

Alternatively, other blockchains, such as Ethereum, serve as the tool of choice for many developers for building decentralized applications.

As compared to Bitcoin, Ethereum has become a more prominent choice because of the flexibility it allows for programming any business logic into the blockchain by using smart contracts.

How to decentralize?

Arvind Narayanan and others have proposed a framework that can be used to evaluate the decentralization requirements of a variety of issues in the context of blockchain technology.

The framework raises four questions whose answers provide a clear understanding as to how a system can be decentralized:

1. What is being decentralized?

Identify what system is being decentralized.

This can be any system, such as a banking or a trading system.

2. What level of decentralization is required?

It can be full disintermediation or partial disintermediation.

3. What blockchain is used?

Determine which blockchain is suitable for a particular application.

It can be Bitcoin blockchain, Ethereum blockchain, or any other blockchain that is deemed fit for the specific application.

4. What security mechanism is used?

How the security of a decentralized system will be guaranteed.

For example, the security mechanism can be atomicity-based, where either the transaction executes in full or does not execute at all.

Other mechanisms may include one based on reputation, which allows for varying degrees of trust in a system.

Blockchain and full ecosystem decentralization

To achieve complete decentralization, it is necessary that the environment around the blockchain also be decentralized.

These elements include storage, communication, and computation.

There are other factors, such as identity and wealth and it is necessary to decentralize these aspects as well in order to achieve a sufficiently decentralized ecosystem.

Storage

Data can be stored directly in a blockchain, and with this fact it achieves decentralization.

However, a significant disadvantage of this approach is that a blockchain is not suitable for storing large amounts of data by design.

It can store simple transactions and some arbitrary data.

It is not suitable for storing images or large blobs of data, as is the case with traditional database systems.

A better alternative for storing data is to use Distributed Hash Tables (DHTs).

DHTs were used initially in peer-to-peer file sharing software, such as BitTorrent and Napster.

BitTorrent is the most scalable and fastest network.

Issue with BitTorrent: There is no incentive for users to keep the files indefinitely.

Two primary requirements here are high availability and link stability,

Availability: data should be available when required.

Link stability: network links should always be accessible.

InterPlanetary File System (IPFS) possesses both of these properties.

IPFS uses Kademlia DHT and Merkle Directed Acyclic Graph (DAG) to provide storage and searching functionality, respectively.

The incentive mechanism for storing data is based on a protocol known as Filecoin, which pays incentives to nodes that store data using the Bitswap mechanism.

The Bitswap mechanism lets nodes keep a simple ledger of bytes sent or bytes received in a one-to-one relationship.

Also, a Git-based version control mechanism is used in IPFS to provide structure and control over the versioning of data.

There are other alternatives for data storage, such as Ethereum Swarm, Storj, and MaidSafe.

Ethereum has its own decentralized and distributed ecosystem that uses Swarm for storage and the Whisper protocol for communication.

BigchainDB is another storage layer decentralization project aimed at providing a scalable, fast, and linearly-scalable decentralized database.

Communication

Access to the internet is based on Internet Service Providers (ISPs) who act as a central hub for internet users.

If the ISP is shut down for any reason, then no communication is possible with this model.

Services such as email and online storage are now all based on a paradigm where the service provider is in control, and users trust such providers.

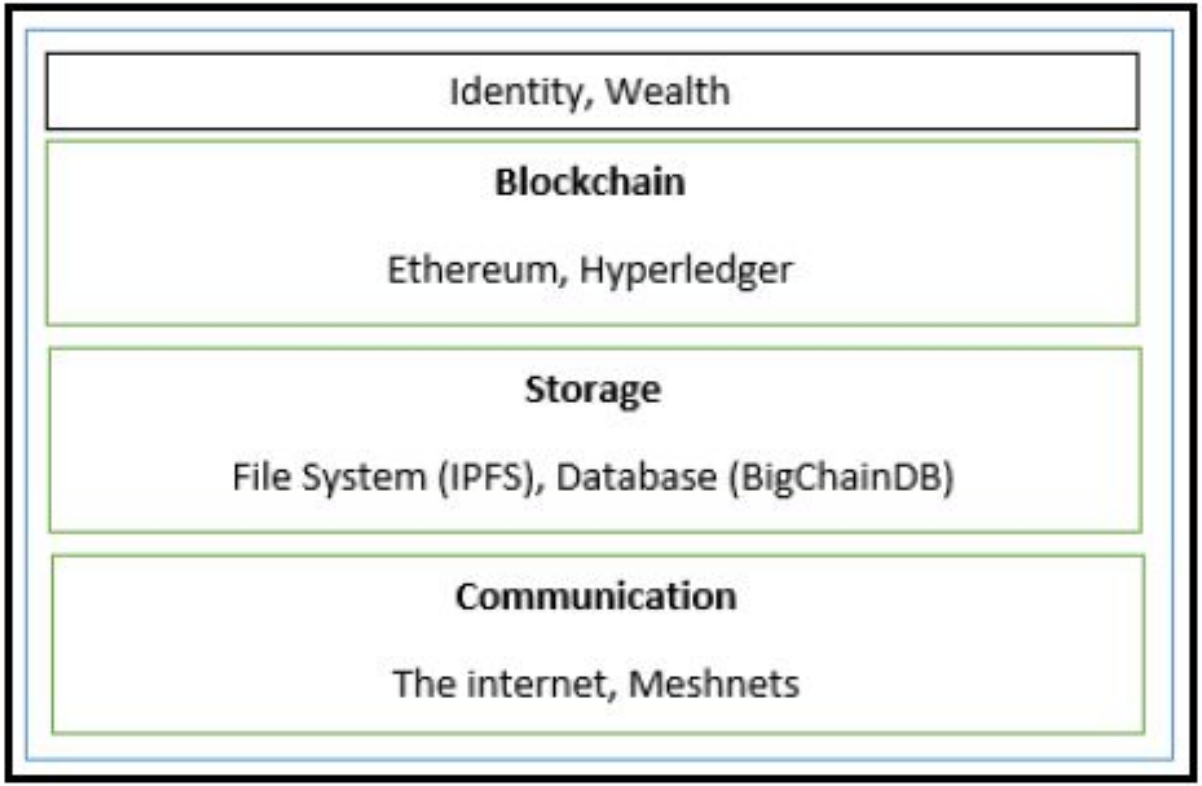
An alternative is to use mesh networks.

They are limited in functionality when compared to the internet, they still provide a decentralized alternative where nodes can talk directly to each other without a central hub such as an ISP.

An example of a Meshnet is FireChat.

Allows users to communicate with each other directly in a peer-to-peer fashion without an internet connection. (Through Bluetooth or WiFi).

A network where no one can shut it down for any reason could be the next step toward decentralizing communication networks in the blockchain ecosystem.



Decentralized ecosystem

Platforms for decentralization

Any blockchain network such as Bitcoin, Ethereum, Hyperledger Fabric, or Quorum can be used to provide decentralization service. Also includes,

Ethereum

Ethereum is the first blockchain to introduce a Turing-complete language and the concept of a virtual machine.

Bitcoin has only a limited scripting language.

Turing-complete language of Ethereum is Solidity.

Endless possibilities have opened for the development of decentralized applications using solidity.

It provides a public blockchain to develop smart contracts and decentralized applications.

Currency tokens on Ethereum are called Ethers.

MaidSafe

MaidSafe provides a Secure Access For Everyone (SAFE) network that is made up of

unused computing resources, such as storage, processing power, and the data connections of its users.

The files on the network are divided into small chunks of data, which are encrypted and distributed randomly throughout the network.

This data can only be retrieved by its respective owner.

Duplicate files are automatically rejected on the network, which helps reduce the need for additional computing resources needed to manage the load.

It uses Safecoin as a token to incentivize its contributors.

Lisk

Lisk is a blockchain application development and cryptocurrency platform.

It allows developers to use JavaScript to build decentralized applications and host them in their respective sidechains.

Lisk uses the Delegated Proof of Stake (DPOS) mechanism for consensus whereby 101 nodes can be elected to secure the network and propose blocks.

It uses the Node.js and JavaScript backend, while the frontend allows the use of standard technologies, such as CSS3, HTML5, and JavaScript.

Lisk uses LSK coin as a currency on the blockchain.

Another derivative of Lisk is Rise, which is a Lisk-based decentralized application and digital currency platform.

It offers a greater focus on the security of the system.